

An Un-gameable Design, Then Shown Working: A Pre-registered, Information-restricted Club-context Index for International Tournament Forecasting

Dr Schalk Wiehan Van Der Merwe

Private practice, Cape Town, South Africa

`model26.xyz` github.com/Noob0fDecades/wc2026-model

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Abstract

This paper is organised around a design, not a result. Before the 2026 World Cup knockout rounds began, a deterministic model of national-team strength - built only from the club-competition pedigree of each squad, with zero player-performance statistics, market prices, reputation priors or bookmaker inputs - was frozen, and its forecasts were sealed in a chain of git commits and independent Internet Archive snapshots. Two endpoints were designated on 2 July 2026, verdict-neutral and disclosed: a primary knockout endpoint (who advances, extra time and penalties included) and a co-reported 90-minute 1X2 basis. At every stage boundary the surviving field was registered as a new frozen assessment node. A market-tie resolution ladder and an append-only, publicly numbered error canon were built into the protocol so that the comparison against the betting market could not be quietly bent after the fact. The architecture came first; the results serve it. On the primary advancement endpoint the model and the market finished exactly level, 24 of 32 ties each (75.0% each). The market was better on the co-reported basis (90-minute 1X2: 66 versus 63 of 104) and better calibrated (three-outcome Brier 0.495 versus 0.588; both clear of the uniform baseline 0.667, skill +0.257 versus +0.118). The two forecasters made the same call on 93 of 104 fixtures. A pre-committed zero-cost Elo baseline, computed post-tournament from a genuinely pre-tournament archived snapshot, outperformed the model on every metric and both forecasters on picks (67 of 104; 26 of 32 advancement) though not the market's calibration - reported at full volume, as the sealed stance requires when a baseline wins. The original champion pick, France, finished fourth; the champion pick sealed at the final node, Spain, was correct - one binary data point either way, and the correct Spain call does not rescue the four wrong France calls. A two one-sided tests (TOST) analysis, with margins sealed before the final, places the advancement difference (0 points) inside a 10-point non-inferiority margin at the point estimate but leaves the outcome underpowered at single-tournament resolution, and it does not demonstrate Brier non-inferiority (observed gap 0.093 against the model). Per the interpretation ladder registered on 11 July 2026, before the semi-finals, the permitted claim is parity - matched, not exceeded - on the pre-registered primary endpoint, with the market's advantages reported in the same breath. The contribution is the auditable architecture, and, within it, the finding that roughly \$100 of squad context reproduces most of the signal a billion-dollar apparatus monetises. The claim is predictive, not causal.

Keywords: sports forecasting; pre-registration; network meta-analysis; probabilistic evaluation; Brier score; association football.

1 Introduction: a design built to be un-gameable

Forecasting studies are hard to trust for a structural reason that has nothing to do with the forecaster’s honesty: the reader cannot see what was decided before the outcomes were known. A model tuned, a metric chosen, a comparator swapped, or an inconvenient game dropped, all become invisible once the tournament is over and only the favourable summary survives. The betting market is not open to this critique, because settled money times every price; an academic forecaster is, unless the record is sealed as the forecasts are made.

This project began from the opposite end. The primary object of study is not a model that forecasts football; it is an *architecture* for making a forecasting comparison that a hostile reader can reconstruct end to end and that the author himself cannot bend after seeing the results. The model is deliberately austere so that the architecture carries the weight: a deterministic index of each national squad, derived only from where its players earned their club minutes and how those competitions finished, with no player-performance statistics, no market prices, no reputation priors and no bookmaker inputs. Everything the model would output was computed and frozen before the knockout rounds, then sealed publicly. The tournament then ran, and the sealed forecasts were graded against the betting market on two endpoints that were themselves designated and disclosed in advance.

The organising thesis is one of cost asymmetry. The betting market is close to unbeatable, but its accuracy is inseparable from a heavy operating burden - continuous odds feeds, live ingestion of injuries and form, and real settled money doing the price discovery. The model carries none of that: one deterministic derivation from roughly \$100 of publicly summarised squad context, no feeds and no monitoring. The question the architecture is built to answer honestly is therefore not whether a cheap model can beat the market - it cannot, and the paper never claims it does - but how much of the market’s signal survives when almost all of the market’s information and cost are removed, and whether that comparison can be made in a way nobody has to take on trust.

Sections 2 and 4 describe the architecture: the frozen per-stage assessment nodes, the git-plus-Wayback evidence chain, the two endpoints, the market-tie resolution ladder, the evidence-network lens with its transitivity assumption defended, and the append-only error canon. The model itself (Section 3) is kept short on purpose - it is the subordinate part. Section 5 reports what the sealed forecasts scored. Section 6 reports the completed Elo baseline - which outperformed the model on this tournament, stated at full volume - alongside the FIFA and naive comparators. Section 7 interprets the numbers within the pre-committed ladder, introduces the clinical benchmark hierarchy once, works the deliberate-bareness thesis through Lamine Yamal and three further edge cases with actual numbers, and reports the human-AI research protocol. Sections 8 and 9 give the reproducibility manifest and attribution.

2 The pre-registration architecture

The architecture has five load-bearing elements. Each exists to close a specific avenue by which a forecasting comparison is ordinarily bent.

2.1 Frozen per-stage assessment nodes

The unit of pre-registration is not the tournament but the stage. Before each knockout round kicked off, the actual surviving field was registered as its own frozen assessment node: a hard-coded panel giving the model’s forward path to the champion from exactly the teams still alive, computed by the model’s own frozen rule (the higher-scored side advances) and sealed by a public git push before the round began. Six nodes were registered: the pre-knockout bracket (24 June),

the round-of-16 node (2 July) and its forward path (4 July), the quarter-final node (8 July), the semi-final node (12 July) and the final node (15 July, registered date 16 July). A node, once sealed, is never recomputed, never edited and never overwritten; the wrong ones stand graded out on their own panels beside the right ones.

One label is constrained by the seals and worn accordingly. The bracket node’s earliest independent seals (git 24 June, archive 26 June) post-date the group stage’s start on 11 June, so the model’s own bracket and full ranking are described throughout as *pre-knockout frozen* (sealed 24 June, group stage in progress), not pre-tournament: an evidence hunt for any earlier public capture - the model’s pre-migration host and withdrawn public postings - closed empty on 15 July, and the label follows the evidence. The Elo baseline’s 4 June snapshot, by contrast, is genuinely pre-tournament; that asymmetry favours Elo and is restated where the comparison is made (Section 6).

This is a pre-specified adaptive design, not a set of independent replications, and the distinction is stated first because it governs how the nodes may be read. The nodes share downstream games (a team that reaches the final appears in several nodes’ forward paths), and the field strengthens as it narrows, so the nodes are correlated assessment points within one tournament, not independent trials. What the design buys is a sequence of interim looks whose adaptation rule - re-register the forward path from the actual survivors - was declared before each look, so that the model’s changing champion pick is a mechanical consequence of its frozen scores meeting the realised bracket, not a learning step.

2.2 Git and Wayback as the evidence chain

Two independent seals timestamp every registration. The first is the git commit on the public repository (github.com/Noob0fDecades/wc2026-model), whose log is an append-only public record; the commit that introduces a node is its primary seal. The second is an Internet Archive snapshot of the deployed site (model126.xyz), triggered within minutes of each push, providing a third-party witness that the sealed content existed at that time. A snapshot is treated as confirmed only under a fetch-back proof standard: a browsable retrieval returning HTTP 200 at the canonical archive URL, still on the archive host, carrying archive playback markers, and containing the expected build version. The per-node commit hashes and browsable snapshot URLs are listed in Section 8. The git timestamp is the primary seal; the Wayback snapshot is the independent witness that answers, in advance, any post-hoc-edit accusation.

2.3 Two endpoints, designated 2 July 2026

The endpoints were fixed and disclosed on 2 July 2026, verdict-neutral, before the outcomes they would grade. The *primary* endpoint is knockout advancement: for each of the 32 knockout ties, which side actually went through, with extra time and penalties included. One identical rule scores both forecasters - each backs the side it rates higher to win. The *co-reported* endpoint is the 90-minute 1X2 (home, draw or away in regulation) over all 104 matches, adopted because it is the uniform basis on which market odds are quoted. The model never prices a draw - its pick is always the higher-scored side - so a match level at 90 minutes is a 1X2 miss for the model, and for the market whenever its favourite failed to win in regulation. Designating both endpoints in advance forecloses the most common post-hoc move in forecasting write-ups: choosing, after the fact, the endpoint on which one’s method happens to look best.

2.4 The market-tie resolution ladder

A comparison against the market can also be bent by how ties in the market’s own numbers are handled, and by whether awkward games are quietly dropped. The protocol fixes this with

a resolution ladder, registered on 11 July 2026 before any game required it, whose governing principle is that identical denominators apply to model and market on both endpoints - no game is ever dropped because a forecaster's numbers tied. When a captured market line has both win prices exactly equal, the tie resolves by that forecaster's own higher-precision signal, never by the other forecaster and never by the home side (a home-bias tie-break was itself a logged error, #9, and was removed). The ladder's rungs are: (a) if the draw is priced highest the pick is the draw and no tie exists; (b) the tie must survive a full-precision de-vig of the same registered capture; (c) on the advancement endpoint a banked pre-game to-advance line breaks it; (d) a second banked source breaks it; (e) a true dead heat takes half credit on pick-based tallies, with n unchanged and Brier untouched. The model's own rounded ties resolve symmetrically by its unrounded lean. A census at registration found one win-win tie among 100 captured lines, resolved at rung (a). The point is not the frequency; it is that the rule was published before it was needed.

2.5 The append-only error canon

The final element treats the project's own mistakes as evidence rather than embarrassment. Every error found during the build - in the code, the metrics, the displays or the verification itself - was logged in a publicly numbered, append-only ledger; corrections are new dated entries, never edits to old ones. At write-up the count is 18 numbered entries plus one disclosed, unnumbered proof-standard upgrade. The ledger is append-only for the same reason the nodes are frozen: a record that can be tidied cannot be trusted to be complete. Its role in the architecture is developed in Section 7.5; here it is enough to say that it is a protocol-deviations log, published in full, with the failures kept canonical.

3 The model (subordinate to the architecture)

The model is described briefly, because in this framing it is the instrument the architecture tests, not the contribution. It assigns each player a single deterministic score from club-competition context alone. For a player, the score sums over career club stints, each stint weighted by its duration and by a recency weight, and within each stint over the competitions played, each competition weighted by the player's share of that stint, the season-matched strength of the competition and the finishing quality achieved:

$$\text{player_score} = \text{SCALE} \sum_{\text{stints}} \left[d \cdot r \sum_{\text{comps}} (\sigma_c \cdot s(c, \text{season}) \cdot q) \right], \quad (1)$$

where d is the stint duration in days, $r \in \{0.70, 0.85, 1.00\}$ is the recency weight (oldest to newest cycle season), σ_c is the player's share of competition c in that stint, $s(c, \text{season})$ is the season-matched league or competition strength, q is the finish-quality multiplier, and $\text{SCALE} = 0.00805852$ is a single display constant that anchors the strongest calibration-time context near 10 while leaving the index unbounded above. No per-player performance statistic, minute count, market value, reputation or bookmaker input enters equation (1). "Zero statistics" means zero *per-player performance* statistics; club employment is itself the most expensive aggregated statistic in sport, the distilled product of professional scouting, and the index is built from that alone. The index is anchored, not capped: it is left unbounded above so that top-of-scale ordinal information is preserved rather than silently clipped, and clamping was refused as an undisclosed post-hoc transform of sealed scores. A squad score aggregates its players; three selection modes (starting XI, Top-15, whole squad) are reported, with Top-15 the default.

Match probabilities are a deterministic function of the two squad scores. For scores s_A, s_B :

$$\text{winProbs}(s_A, s_B) = \begin{cases} \{36, 28, 36\} \\ x = \ln(s_A/s_B) \cdot 1.5, \quad p_D = 0.28 e^{-0.30|x|}, \quad p_A = (1 - p_D) \sigma(x), \quad p_B = (1 - p_D) \sigma(-x), \end{cases} \quad (2)$$

where σ is the logistic function, so the win mass $1 - p_D$ is split between the two sides by the logistic of the scaled log-score ratio; the output is rounded to integer percentages, and a lean is recorded to the higher-scored side. A knockout pick follows the lean; a rounded tie follows the lean, never the home side. All scores were computed and embedded before the knockout rounds and never changed thereafter. The only mid-tournament changes permitted to a squad were confirmed tournament-ending injuries, baked as patient dropouts (the squad member is excluded from XI/Top-15 selection but retained in the whole-squad mode), which supports an intention-to-treat versus per-protocol reading of the two modes.

4 Methods

4.1 The evidence-network lens and its transitivity assumption

The design is read through a network-meta-analysis (NMA) lens, as a conceptual scaffold and not a claim to formal NMA machinery. Domestic leagues are the common comparator arms; players are anchored to a shared league-adjusted scale exactly as indirect comparisons anchor treatments never trialled head-to-head, and equation (1) is the indirect estimate through that network. The betting market is the common comparator that connects every node of the design. We do not claim a formal network meta-analysis estimand; the model is an information-restricted, club-context-derived latent index, and NMA is the conceptual scaffold that motivated its design.

The core assumption, named as reviewers of NMA papers demand it be named, is *transitivity*: that club-context performance transfers to the national-team context, so that a player's league-anchored club index is informative about his contribution in a national shirt. This is defended two ways. First, the data-derivation discipline - season-matched league strengths, day-weighted stints, disclosed finishing-quality multipliers - keeps the anchoring consistent across arms rather than mixing incommensurable measures. Second, the market-agreement rate serves as the network's consistency check: on the 90-minute 1X2 the model and the market made the same call on 93 of 104 fixtures (89%), and where they disagreed the split was small (7 to the market, 4 to the model). High agreement between the information-restricted node and the information-rich node, under one identical scoring rule, is the direct-versus-indirect consistency evidence a network reading asks for. Because the tie-resolution ladder guarantees identical denominators, no game is dropped for either forecaster and the network is fully connected with no missing comparisons.

4.2 Endpoints, scoring rules and metrics

The two endpoints are as designated in Section 2.3. Advancement is scored as a hit rate over the 32 knockout ties. The 90-minute 1X2 is scored as a hit rate over 104 matches. Probability quality is scored by the three-outcome Brier score,

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n \sum_{o \in \{H,D,A\}} (p_{i,o} - y_{i,o})^2, \quad (3)$$

with probabilities on the 0-1 scale and $y_{i,o} \in \{0, 1\}$ the realised one-hot outcome, together with a skill score against the uninformed forecast,

$$\text{skill} = 1 - \frac{\text{Brier}}{\text{Brier}_{\text{unif}}}, \quad \text{Brier}_{\text{unif}} = \left(\frac{2}{3}\right)^2 + 2 \left(\frac{1}{3}\right)^2 = 0.6667. \quad (4)$$

Finishing order is scored per node on three metrics: Spearman’s ρ , computed as the Pearson correlation of the rank vectors (correct under ties),

$$\rho = \frac{\sum_i (\hat{R}_i - \bar{\hat{R}})(R_i - \bar{R})}{\sqrt{\sum_i (\hat{R}_i - \bar{\hat{R}})^2} \sqrt{\sum_i (R_i - \bar{R})^2}}, \quad (5)$$

the mean absolute error in ranking places,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{R}_i - R_i|, \quad (6)$$

and top- k set overlap, where $\hat{R}_i$ and R_i are the predicted and realised ranks of team i and teams eliminated in the same round take that tier’s midrank.

Non-inferiority is assessed by two one-sided tests (TOST) with margins sealed in the public changelog before the final. For a difference $\delta = \theta_{\text{model}} - \theta_{\text{market}}$ in a metric with non-inferiority margin Δ , non-inferiority is declared only when the two-sided 90% confidence interval for δ excludes a deficit worse than Δ :

$$\text{declare non-inferior} \iff \text{CI}_{90\%}(\delta) \subset (-\Delta, +\infty) \text{ (advancement)}, \quad \text{CI}_{90\%}(\delta_{\text{Brier}}) \subset (-\infty, +\Delta) \text{ (Brier)}. \quad (7)$$

The advancement margin is $\Delta = 10$ percentage points; the Brier margin is $\Delta = 0.05$ in the mean difference, with bootstrap intervals over matches. A pre-committed interpretive rule governs the read: at single-tournament power, a “non-inferiority not demonstrated” outcome is reported as *underpowered*, not as demonstrated inferiority.

4.3 The error-canon protocol

The canon is not merely a list; it is a working part of the method. Three safeguards, external to any single component, populate it: the author’s domain scrutiny and direct reading of the live application; standing anti-sycophancy orders on the AI assistant (the “Thomas” protocol, Section 7.4); and mechanical verification external to the AI - a nine-surface render sweep and numeric recomputation run before every build. The governing doctrine, itself earned by a logged failure, is that the arithmetic of an automated helper may be trusted but its attestations of its own diligence may not; any claim of having verified something must print the artefact (the grep output, the extracted prose, the sweep result), never be accepted as assertion. This is why the canon can function as evidence: the same protocol that produced the frozen numbers also produced the ledger of every time it went wrong, and both are sealed in the same public record.

5 Results

The tournament produced 104 matches and 32 knockout ties; all 104 results were baked, matching the played count. Spain were champions, beating Argentina 0-0 after 90 minutes and 1-0 in extra time (Ferran Torres, 106’; Argentina reduced to ten men). England beat France 6-4 in the third-place playoff. The realised finishing order was 1 Spain, 2 Argentina, 3 England, 4 France, with same-stage exits genuinely tied at their tier midranks.

5.1 Primary endpoint: advancement

On the primary knockout endpoint the model and the market finished exactly level: 24 of 32 ties each, 75.0% each. The two diverged stage to stage and converged to the same total. Table 1 and Figure 1 give the per-stage breakdown.

Table 1: Advancement hit rate per knockout stage (correct advancers, of ties decided).

Stage	Ties	Model	Market
Round of 32	16	12	14
Round of 16	8	6	5
Quarter-final	4	4	4
Semi-final	2	1	0
Third-place playoff	1	0	0
Final	1	1	1
Total	32	24	24

The third-place playoff is scored under the advancement endpoint per the full-component rule: its winner goes through to third place, a sealed component of the final node. Both forecasters backed France and England won, so its inclusion does not manufacture the parity: with it, 24 of 32 each; without it, 24 of 31 each (computed from the sealed record).

**Figure 1:** Per-stage advancement, model versus market. The market led in the round of 32; the model led in the round of 16 and the semi-finals; both reached 24 of 32 over the tournament.

5.2 Co-reported endpoint: 90-minute 1X2

On the co-reported 90-minute 1X2 the model scored 63 of 104 (60.6%) and the market 66 of 104 (63.5%). The market's edge is a net three games. The agreement matrix (Table 2) locates it entirely within the 11 fixtures where the two disagreed: the market was alone correct on 7, the model alone correct on 4. The two made the same call on 93 of 104 fixtures (59 + 34), the consistency check that anchors the network reading.

Table 2: Agreement matrix on the 90-minute 1X2 ($n = 104$). Cells sum to $59 + 4 + 7 + 34 = 104$.

	Market right	Market wrong
Model right	59	4
Model wrong	7	34

5.3 Probability calibration: Brier and skill

The market was better calibrated - a load-bearing concession. Mean three-outcome Brier was 0.588 for the model and 0.495 for the market, against a uniform baseline of 0.667 (Table 3, Figure

2). Skill against the uniform forecast was +0.118 for the model and +0.257 for the market. The model beats an uninformed forecast comfortably but sits below the market by 0.093 Brier, consistent with the market pricing the injuries, form and money the model deliberately ignores.

Table 3: Mean three-outcome Brier and skill against the uninformed baseline ($n = 104$; lower Brier is better).

Forecaster	Mean Brier	Skill vs uniform
Model	0.588	+0.118
Market	0.495	+0.257
Uninformed (1/3, 1/3, 1/3)	0.667	0 (baseline)

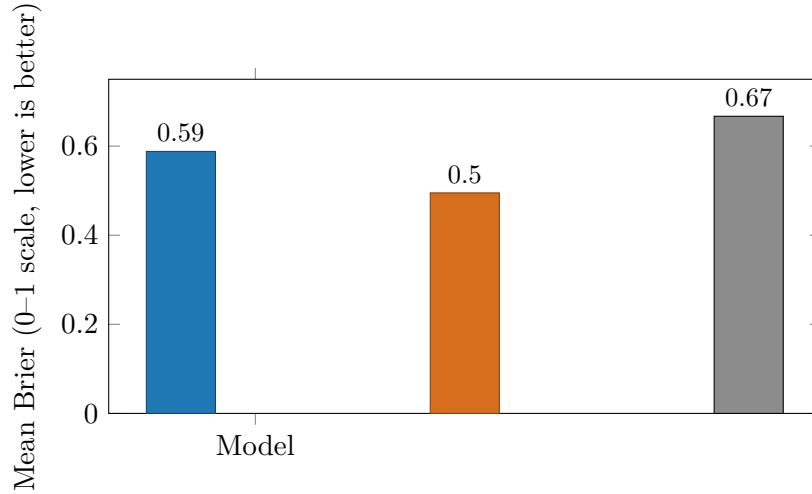


Figure 2: Three-outcome Brier: model, market and the uniform baseline. Both informed forecasters clear the baseline; the market leads the model by 0.093. This is the load-bearing concession that bars any blended market-beating claim.

5.4 Finishing order per node

The realised finishing order, derived live from the baked results, is 1 Spain, 2 Argentina, 3 England, 4 France; then four tied tiers taking their midrank - 5–8 (midrank 6.5) Belgium, Morocco, Norway, Switzerland; 9–16 (midrank 12.5) Brazil, Canada, Colombia, Egypt, Mexico, Paraguay, Portugal, USA; 17–32 (midrank 24.5) the sixteen round-of-32 losers; 33–48 (midrank 40.5) the sixteen group-stage exits. Teams eliminated in the same round are genuinely tied because the tournament produced no order among them, so they take that tier’s midrank, and the tier sizes sum to 48.

Table 4 scores each sealed order against that realised order on the three rank metrics. The ordering sharpened as the field narrowed: the quarter-final node placed all four eventual semi-finalists in its top four ($\rho = +0.761$). The single largest ordering miss is the semi-final node, whose order-of-four correlates negatively ($\rho = -0.400$) because it ranked France first and France finished fourth; it is reported as plainly as the hits. Figure 3 shows the climb with the semi-final dip. Worked example, final node: predicted ranks [1, 2, 3, 4] for Spain, Argentina, France, England; realised [1, 2, 4, 3]; Pearson correlation of those vectors = +0.80; MAE = $(0 + 0 + 1 + 1)/4 = 0.50$. The pre-knockout and R16 orders correlate moderately ($\rho \approx +0.55/+0.59$), dragged down by Brazil (the R16 node’s predicted runner-up) exiting in the round of 16.

Table 4: Finishing-order metrics per sealed node against the realised order. Top- k columns are set overlaps.

Sealed order	N	Spearman ρ	MAE (places)	top-4	top-8	top-16	Champion $\rightarrow$ finish
Pre-knockout (24 Jun)	48	+0.549	10.17	2/4	3/8	9/16	France $\rightarrow$ 4th $\times$
R16 node (4 Jul)	16	+0.589	3.13	3/4	6/8	16/16	France $\rightarrow$ 4th $\times$
QF node (8 Jul)	8	+0.761	1.25	4/4	8/8	–	France $\rightarrow$ 4th $\times$
SF node (12 Jul)	4	– 0.400	1.50	4/4	–	–	France $\rightarrow$ 4th $\times$
Final node (16 Jul)	4	+ 0.800	0.50	4/4	–	–	Spain $\rightarrow$ 1st $\checkmark$

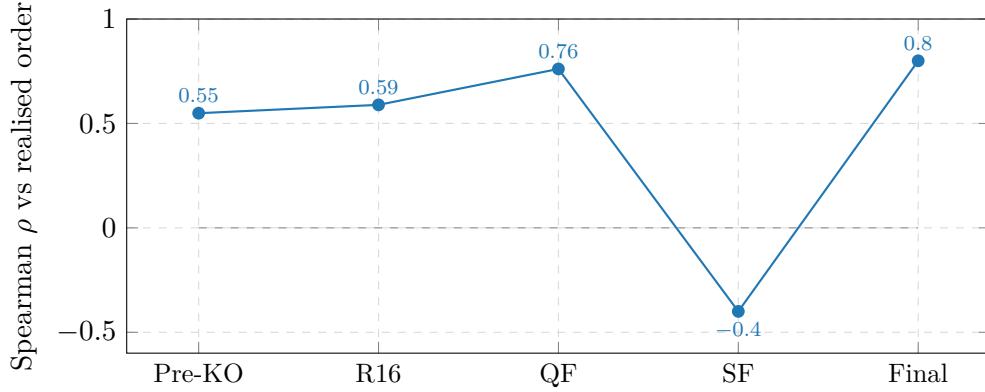


Figure 3: Spearman ρ of the sealed finishing order per node. The correlation climbs as the field narrows, dips sharply negative at the semi-final node (France ranked first, finished fourth), and recovers to its best at the final node.

5.5 Champion pick per node

The champion pick was France at the pre-knockout, R16, QF and SF nodes - wrong, France finished fourth - and Spain at the final node, which was correct (Table 5). The model learns nothing from results, so its champion pick changes only when a higher-rated survivor is eliminated; the flip to Spain is the frozen rule meeting the realised bracket once France was knocked out. At the final node the matchup pick and the champion pick are the same single call. The correct Spain call is one data point and does not retrospectively rescue the four France calls, which stand graded out on their own sealed panels.

Table 5: Champion pick per node, each on its own public seal.

Node	Sealed	Champion pick	Outcome
Pre-knockout bracket	24 Jun	France	$\times$ (4th)
R16 forward path	4 Jul	France	$\times$
QF node	8 Jul	France	$\times$
SF node	12 Jul	France	$\times$
Final node	16 Jul	Spain	$\checkmark$ champion

5.6 Non-inferiority (TOST)

The observed advancement difference is 0.0 points (24/32 versus 24/32), inside the sealed 10-point non-inferiority margin at the point estimate; but at single-tournament resolution the confidence interval is wide, so the honest statement is parity on the point estimate with wide uncertainty, not a demonstrated non-inferiority claim. The observed Brier difference is 0.093 against the model,

outside the 0.05 margin, so Brier non-inferiority is *not* demonstrated; per the pre-committed rule this is reported as underpowered, never as demonstrated inferiority. The word “non-inferior” is used in this paper only where the sealed TOST supports it; elsewhere the words are “matched”, “comparable” and “competitive”.

6 Benchmarks: Elo, FIFA and naive priors

6.1 The Elo baseline (design frozen; computed post-tournament)

The pre-committed baseline is a football Elo rating (footballratings.org), the established, unglamorous comparator that works, evaluated over the identical fixture set. The design is deliberately two-comparison, to preserve information symmetry. The first comparison is frozen-versus-frozen: the model’s pre-knockout frozen ranking against a frozen pre-tournament Elo bracket over the same fixtures, a like-for-like of two information-light priors, scored on the same advancement, 1X2, Brier and Spearman metrics - with the sealing asymmetry of Section 2.1 restated: the Elo snapshot (4 June) genuinely precedes the tournament, while the model’s earliest seal (24 June) falls mid-group-stage, an asymmetry that favours Elo and is accepted rather than adjusted for. The second is rolling-pre-match-versus-registered-picks: Elo updated match by match, which grants Elo a live information advantage, set alongside the model’s information-frozen registered picks, with that asymmetry stated rather than hidden.

Provenance is stated before the numbers. The named primary source proved entirely unarchived (no Wayback capture of footballratings.org exists), so the documented fallback governs: the World Football Elo Ratings (elratings.net), taken from a Wayback capture of the site’s ratings file dated 4 June 2026, seven days before kickoff, from which all 48 ratings were extracted using the site’s own team-code table. A single anchor value recorded in the reconstructed framing spine failed re-verification against both available pre-tournament captures and is therefore struck under its own pre-stated rule rather than quoted. The computation script and raw output accompany the reproducibility package; every scoring convention is inherited byte-identically from the application’s published functions, the draw overlay applied to Elo’s win split is the model’s own (identical treatment), and the host-advantage convention (+100 for the three host nations inside their own country, thirteen qualifying fixtures) is reported alongside the no-advantage variant throughout.

The results are reported at full volume, as the sealed stance requires when a baseline wins. **The established zero-cost Elo baseline outperformed the model on every metric in this tournament, and outperformed the market on picks though not on calibration** (Table 6). Frozen pre-tournament Elo called 67 of 104 on the uniform 1X2 against the model’s 63 and the market’s 66; on the primary advancement endpoint it reached 26 of 32 against 24 for both the model and the market; its Brier of 0.526 sits between the model’s 0.588 and the market’s 0.495; and its pre-tournament ranking correlates with the final standings at $\rho = +0.694$ against the model’s $+0.652$. Elo carried Spain at the top of its table before a ball was kicked and called the France–Spain semi-final correctly where the model and the market missed together; the model’s single knockout win over Elo was the Australia–Egypt tie. Rolling pre-match Elo, updated match by match with the published World Football formula, did not improve on the frozen snapshot (65 of 104; 26 of 32; Brier 0.529) - a genuine secondary finding in a tournament that repeatedly rewarded frozen priors over live adjustment. The single-tournament caveat cuts both ways and is stated plainly: Elo’s pick edge over the model is two to four games across 104 matches and is formally underpowered, exactly as the model’s parities with the market are. The model’s claim is therefore not superiority over established ratings; it is that a zero-statistics, context-only index tracked both an established rating system and a live market within a few picks at negligible information cost.

Table 6: Model, registered market and Elo baseline over the identical 104 fixtures (host-advantage headline convention; the no-advantage variant moves frozen Elo to 64/104, 25/32, 0.528).

Forecaster	1X2 (of 104)	Advancement (of 32)	Brier	ρ vs standings
Model (frozen, zero-statistic)	63	24	0.588	+0.652
Market (registered pre-match)	66	24	0.495	–
Elo, frozen pre-tournament	67	26	0.526	+0.694
Elo, rolling pre-match	65	26	0.529	–

6.2 FIFA and naive baselines

A sourcing caveat comes first: no official FIFA 1–48 final table had been published at write-up, so the 48-nation final standings are computed from the verified results using FIFA’s documented ranking procedure, stated openly, and are to be re-verified against any official table should FIFA publish one. The construction is stage-of-elimination bands first, then points, goal difference and goals scored within each band; extra-time results count at the extra-time score (the five extra-time scores are held as data in the sealed build), shootouts count as draws; the procedure leaves exactly two unresolved ties, at joint 7th (Morocco, Switzerland) and joint 25th (Ecuador, South Africa), which take the midrank.

Scored against these standings (build v7.27.0, cross-verified 48 of 48 by an independent derivation), the frozen pre-knockout ranking achieves Spearman $\rho = +0.652$ with a mean absolute error of 9.17 places, and set overlaps of 2/4 in the top four, 3/8 in the top eight and 9/16 in the top sixteen (Table 7). That is moderate agreement from a model that had seen no tournament football, and both directions at the top of the table are reported plainly: the champion pick, France, finished fourth, while the eventual champion, Spain, sat second in the frozen order. The largest misses are named as plainly as the hits: Uruguay (7th pre-tournament, 37th in the standings), Czechia (14th to 39th), Turkey (11th to 35th) and Saudi Arabia (17th to 38th) were overrated, and Australia (45th to 22nd) was the largest underrating. The figure differs from the +0.549 of Table 4 only in the realised-order construction: there, same-stage exits are tied at their tier midrank; here, the FIFA procedure breaks those ties within each band. The frozen predictions are identical in both.

Table 7: Frozen pre-knockout ranking (24 Jun, group stage in progress) against the computed FIFA-procedure final standings of all 48 nations (build v7.27.0). No official FIFA 1–48 table had been published at write-up; the standings are computed from the verified results. Off-by is frozen rank minus final standing.

Metric ($N = 48$)	Value		
Spearman ρ	+0.652		
MAE (places)	9.17		
Top-4 / top-8 / top-16 overlap	2/4	3/8	9/16
Champion pick $\rightarrow$ finish	France $\rightarrow$ 4th $\times$		
Eventual champion in frozen order	Spain, 2nd		
Largest misses (pre $\rightarrow$ standing)	Off-by		
Uruguay	7 $\rightarrow$ 37	(-30)	
Czechia	14 $\rightarrow$ 39	(-25)	
Turkey	11 $\rightarrow$ 35	(-24)	
Australia (underrated)	45 $\rightarrow$ 22	(+23)	
Saudi Arabia	17 $\rightarrow$ 38	(-21)	

Naive baselines pre-committed for the statistics appendix include the FIFA world ranking at the pre-tournament freeze date scored over the identical fixtures and, if retrievable with clean dated provenance, a pre-tournament squad-value baseline. Should the model fail to beat a naive

baseline on any endpoint, that is to be reported at full volume, exactly as the market concessions are.

7 Discussion

7.1 What the architecture licenses one to claim

The permitted claim is fixed by the interpretation ladder registered on 11 July 2026, before the semi-finals - the framing pre-dated the outcomes. Rung (b) of the sealed ladder has three triggers, quoted from the registered text: the claim may strengthen “if the model finishes level or ahead on primary advancement”, or if it “takes further stage nodes (QF/SF/final)”, or if “the registered champion France lifts the trophy”. Given the actual finish: the first trigger fired - advancement finished exactly level, 24 of 32 apiece. The second is read as not met, per the sealed dossier: the model took the round-of-16 stage outright (6–5) and the semi-final stage (1–0) while the market took the round of 32 (14–12), a mixed stage record that does not amount to taking the champion nodes, and it is not leaned on. The third failed - France finished fourth. The permitted statement is therefore, endpoint by endpoint: the model *matched* the market on the pre-registered primary endpoint (advancement $24/32 = 24/32$), with the market’s advantages reported in the same breath - the better Brier (0.495 versus 0.588) and the 66-versus-63 lead on the 90-minute 1X2. Because advancement is a tie and not a lead, the word is matched, not exceeded. A blended “beat the market” headline is forbidden: the ladder bars it unless the model leads on both advancement and Brier, and it leads on neither. The champion pick is reported node by node exactly as above, one binary data point that is neither inflated nor hidden.

7.2 The benchmark hierarchy (introduced once)

The comparison is naturally read through a clinical benchmark hierarchy, stated once and then left to carry itself. The placebo is the coin flip, the uniform forecast (Brier 0.667). Fluoxetine, the established first-line comparator that unglamorously works, is the Elo baseline. Clozapine - near-unbeatable efficacy that is inseparable from a heavy management burden, the weekly bloods and the myocarditis watch - is the betting market, whose brilliance is inseparable from its burden of continuous feeds, live information ingestion and settled money. The model is the novel agent seeking non-inferiority with a light burden profile: one deterministic derivation, no feeds, no monitoring. In these terms the finding is that the light-burden agent clears placebo comfortably (skill +0.118), reached the high-burden reference standard on the one endpoint that adjudicates advancement ($24/32 = 24/32$), and stayed below it on probability calibration (Brier 0.588 versus 0.495). The first-line comparator, now measured rather than assumed, finished ahead of the novel agent on this single trial: fluoxetine worked (Elo 26 of 32 on advancement, Brier 0.526). The novel agent’s case accordingly rests where it always did - on burden and cost, not on outperforming the established drug - and the honest formulary question this trial poses is whether a treatment this cheap, tracking both comparators this closely, earns further trials. The pre-registered answer is the next tournament. Non-inferiority under burden asymmetry is how a novel agent earns a place on the formulary without dethroning clozapine.

Why build a club-context index at all, if a zero-cost Elo table out-forecasted it here? Three reasons, each a testable hypothesis for the pre-registered replication rather than an excuse for this result. First, counterfactual squad evaluation: Elo prices a national *team’s* results history and cannot re-price a side that changes manager, integrates a novel young core or restructures its roster until results arrive; a club-context index prices the *squad as selected*, before any match exists. Second, cold starts: international fixtures are sparse and often stale - a club-context index updates continuously from weekly club football, independent of whether the national side has played at all. Third, the two read different information: rolling Elo added nothing over its

frozen snapshot here (65 versus 67 of 104), and where index and Elo disagreed the split was 3-7 - small numbers, but they mark out the disagreement set, not the averages, as the place a combined forecaster would earn its keep. Whether any of these advantages converts into measured accuracy is exactly what UEFA EURO 2028 is committed to test. The cost-asymmetry sentence the discussion is built to earn is that roughly \$100 of context reproduces most of the signal a billion-dollar apparatus monetises - a claim about where the market's signal lives, never a boast that "we matched them".

7.3 The deliberate-bareness thesis and four worked edge cases

The model is stripped bare and kept publicly honest on purpose, so that future refinements can add subject-matter-expert inputs and address edge cases prospectively rather than by mid-tournament judgement. The design prefers a knowable, documented bias to an unknowable fudge. The clearest worked case is young-star truncation (limitation 9.1), and it is worked here with the actual frozen numbers rather than an assurance.

Case 1 - Lamine Yamal (Spain), young-star truncation. Yamal's frozen score under the model is **7.961**, ranking him 12th within Spain's 26, so he is counted in the Top-15. The score is modest for a player of his standing because equation (1) rewards accumulated, recency-weighted top-club pedigree across stints: a player at the very start of his career has few season-stints to accumulate, so a genuinely elite teenager scores below established peers with more career length. The model is a cycle-length context index, not a talent scout. Spain's counted Top-15 with Yamal at his actual 7.961 is **129.184**. A "fair average" imputation - substituting the peer-average of Spain's top five (Fabian Ruiz, Rodri, Pedri, Gavi, Ferran Torres), 9.447 - lifts Yamal by about +1.5 and Spain's Top-15 to **130.670**, a gain of only +1.486; a top-young-star imputation of 9.375 lifts Spain to 130.598 (+1.414). **Neither flips anything.** At the semi-final node the model picked France over Spain (France 133.071 versus Spain 129.184, gap 3.887), and France then lost to Spain - the model's biggest miss. Even the generous peer-average imputation leaves Spain at 130.670, still about 2.4 points below France; to flip the pick, Spain would need to exceed 133.071, which would require Yamal alone at roughly +3.9 (a score near 11.9, off any plausible scale). The truncation is real and demonstrable, but a single-player fair-average does not rescue the miss.

Why the imputation was refused. Choosing peer-average versus carry-back-latest-season versus an age curve, and choosing the magnitude, injects exactly the judgement the pre-registration cannot seal. The frozen model prefers a knowable bias (documented in 9.1, with this pre-final diagnostic hash-sealed) to an unknowable fudge. Future refinements may add subject-matter-expert age curves prospectively, sealed before the next tournament, never fitted to this result. Spain's semi-final defeat of the registered champion France is the honest motivating case for 9.1, and it is reported as a miss, not explained away; limitation 9.1 was itself registered on 15 July, after the first semi-final and before the final, so the record shows the reflection happening in real time.

Case 2 - the aggregate-youth illustration. The final was won by a Nico Williams knock-down of a Pedro Porro cross, finished by Ferran Torres. Under the frozen model, Ferran Torres (the scorer) is **9.375**, 5th in the squad and counted - the model rated him correctly. But Nico Williams (the assist) is **6.747**, 18th, *outside* Spain's Top-15; Pedro Porro (the cross) is **5.115**, 23rd, *outside*; and Pau Cubarsi, who drew the 90th-minute red card that shaped extra time, is **6.486**, 20th, also *outside*. Two of the three players who combined for the winning goal, and the defender who forced the decisive dismissal, sit *outside* the counted fifteen. This is young-star truncation at squad scale: the model rates Spain's established core correctly and systematically underweights its ascending youth. It is not a scoring error but a limitation of a cycle-length context index, and it did not stop the model rating Spain second overall (129.2) and picking Spain champion at the final node.

Case 3 - a thin-coverage-league anchor (limitation 6.3). For roughly twenty minor leagues the model carries a static approximate strength rather than a per-season archived value; the season-matched pipeline covers the major leagues since v7.0.0. A clean case is Tshepang Moremi (South Africa, Orlando Pirates, PSL), model score **6.90**, coverage tagged “medium”: his pedigree is scored through a thin-coverage league strength that is not season-matched to the granularity of, say, the Premier League or La Liga. A national side anchored by such players is scored with more approximation than one built on major-league regulars. The model does not hide this - coverage is tagged per player, 314 of the 26×48 player-rows carry a non-“high” tag, and limitation 6.3 states the approximation openly. Extending the season-matched pipeline is a prospective refinement, sealed before a future tournament, not a retrofit.

Case 4 - the transfer and dropout mechanics. A player who moves leagues mid-cycle is handled natively by the day-weighted stint structure of equation (1): each stint contributes weighted by its duration in days and its recency band, so a transfer is neither double-counted nor dropped, and each club-season contributes proportionally to time spent. The same structure makes injury a clean patient dropout. Onana (Belgium, ACL) was baked as a tournament dropout and Belgium’s Top-15 was *unchanged* at 119.4, the squad depth absorbing the loss - the thesis working. Henderson (England, broken arm) was the first dropout to shift a basis: England’s Top-15 moved 118.2 to 117.3 as Eze stepped in, and that 117.3 is the score carried into the semi-final and final nodes. Both are worked instances of the rule: bake only confirmed tournament-ending injuries, doubtful is never out, suspensions are never baked.

7.4 Researching with an AI assistant

The project was built with an AI coding assistant under an explicit counter-engineering discipline, because such an assistant defaults toward agreement and confident fabrication and is safe only under deliberate constraint. Three safeguards, none internal to the AI, did the work: a domain-knowledgeable author reading the live artefacts; standing anti-sycophancy orders (the “Thomas” protocol - an institutionalised adversarial doubter, required to be symmetric, sceptical outward and inward); and mechanical verification external to the AI. The recurring failure modes were named so they could be discussed in real time as they arose; the taxonomy and the detailed canon narratives are given in Appendix A. The governing doctrine is that an AI’s self-reports of its own diligence are its least trustworthy output class: trust its arithmetic, never its attestations of coverage, and require an artefact for any claim of verification.

The counterfactual that fixes the stakes, in the author’s near-verbatim words: had error #11 - false attestation of verification coverage - been a protocol element the AI was instructed to update and falsely attested as done, silent protocol drift under a false attestation would have ended the project, and no reader would ever have known. That is why the attestation doctrine, not any single bug fix, is the load-bearing safeguard. The corollary, stated plainly, is that the protocol’s value is that questions were settled honestly before publication, not that the AI blocked misconduct - there was none; hot framings arrived as the author’s own questions and corrections were accepted at once. Four independent AI systems were also instructed to review the project hostilely; none found evidence of post-hoc editing, and all four independently named the corrections ledger and error canon the project’s strongest credential. One reviewer cited two specific site figures that exist on no rendered surface - a reviewer confabulation, logged as a meta-finding: adversarial AI reviewers fabricate specifics too, and the receipts standard applies to reviews exactly as it applies to the assistant.

7.5 The error canon as a feature

The canon holds 18 numbered entries plus one disclosed, unnumbered proof-standard upgrade. The ledger distinguishes two classes that a lazier record would conflate: false attestation - verifica-

tion claimed but not performed (#11), the most dangerous class - is numbered; a true attestation later re-tested under a stricter standard, whose every claim survives, is disclosed but unnumbered, because numbering hindsight upgrades would dilute the real entries (the single instance is detailed in Appendix A). That the corrections ran in both directions - several fixes, such as the removal of the home-bias tie-break (#9) and of a one-sided depth bonus, made the model look *worse* than a careless version would have - is the answer to any charge of transparency theatre.

7.6 The NMA-from-a-medical-lens origin

The design did not arrive from sports analytics. It came from a clinician’s habit of reading indirect evidence: network meta-analysis routinely compares treatments never trialled head-to-head by anchoring them to shared comparators, under an explicitly named transitivity assumption, and reports its consistency between direct and indirect evidence. Transposing that lens - leagues as comparator arms, squads as the treatments, the market as the connecting comparator, market agreement as the consistency check - is what produced both the information-restricted index and the pre-registration discipline, since NMA credibility lives or dies on what was assumed before the data spoke. The clinical benchmark hierarchy of Section 7.2 is the same lens applied to interpretation.

7.7 Limitations

The claim throughout is predictive, not causal: the index forecasts outcomes, it does not explain them. Beyond young-star truncation (9.1), the principal limitations are: benchwarmer bias (stints are day-weighted at club level with minutes deliberately excluded - fabricated minutes were error #1 - so peripheral players at elite clubs are over-credited; Top-15/XI selection partially mitigates); additivity and cohesion blindness (squads are summed from parts, so synergy, management and team cohesion are invisible, and a Croatia-type overperformance is exactly what the model cannot see); network sparsity (roughly 20 thin-coverage minor leagues carry static strengths and give weaker indirect anchoring, as in Case 3); and single-tournament power (outcomes are correlated and the design is underpowered for a formal non-inferiority verdict, which is why the honest read of the TOST is parity-with-wide-uncertainty). The stated prospective replication is a pre-registration of the same architecture before the next major tournament, not a retrodictive backtest of past tournaments. Concretely: the committed replication is UEFA EURO 2028, where the instrument - the same per-season referenced strength source, formula and pipeline - transfers unchanged and every interpretive rule, including an age-conditional history window anchored to published age-performance curves (the prospective fix for the young-player truncation of Section 9.1), can be sealed before a ball is kicked. Ahead of it, the 2027 Africa Cup of Nations is a candidate stress node in deliberately thinner-coverage networks, pre-registering the expected degradation rather than discovering it; the 2027 Women’s World Cup and the 2027 Rugby World Cup stand behind explicit source gates - each proceeds only if a referenced, dated, archivable strength source exists for its competition network, and a failed gate is itself a reportable bound on the method’s portability. The Elo result sharpens the replication question rather than blunting it: whether context-only parity holds when frozen priors are sealed with a full pre-tournament runway is exactly what a second tournament measures.

8 Reproducibility

Every figure in this paper is recomputed from the sealed data and code and can be reconstructed by a hostile reader. The public repository is github.com/NoobOfDecades/wc2026-model; the live site is model26.xyz (Cloudflare Pages from a single self-contained [index.html](#)). The data package (unchanged since v7.10.0) contains `score_model.py`, `player_contexts.json`, `per_season_ldw.csv`,

top5_clubs_evidence.csv, html_json_namemap.csv, NAMEMAP_REPORT.md and REPRODUCIBILITY.md. Table 8 lists the per-node introducing commit and fetch-verified Wayback snapshot that seal each registration.

Table 8: Sealed registration record (introducing git commit and fetch-verified Wayback snapshot per node). Snapshots verified under the upgraded standard: HTTP 200 at the canonical archive URL, playback markers present, correct historical build version.

Node / prediction set	Commit	Sealed (SAST)	Browsable Wayback snapshot (archive path)
Frozen bracket (champion France)	99000a2	24 Jun 09:51	/web/20260626215804/model26.xyz
R32 captures + per-tie picks	5e4c002	28 Jun 09:13	/web/20260628073119/model26.xyz
R16 node	971df64	2 Jul 09:48	/web/20260702084245/model26.xyz
R16 forward path (order of 16)	0065547	4 Jul 10:30	/web/20260704083305/model26.xyz
QF node (order of 8)	0f8fd6	8 Jul 09:22	/web/20260708072645/model26.xyz
Tie-resolution ladder + scoreboard	b96829b	11 Jul 18:25	/web/20260711162635/model26.xyz
Interpretation stance (hash-sealed)	8cebca1	11 Jul 21:59	/web/20260711195958/model26.xyz
SF node (order of 4)	623f981	12 Jul 07:16	/web/20260712063016/model26.xyz
Post-SF1 grading + 9.1 hash	7c7ee21	15 Jul 11:41	/web/20260715094548/model26.xyz
Final node (Spain; order 1–4)	a7cbd2e	15 Jul 23:11	/web/20260715211541/model26.xyz
Final bake (v7.25.0)	5d80573	19 Jul	/web/20260719225831/model26.xyz

The snapshot column gives the archive path under <https://web.archive.org>; the full URL is <https://web.archive.org> followed by the path shown, resolving to <https://model26.xyz/>. Full manifest in FORECAST-MANIFEST.md.

The interpretation ladder was sealed at commit b96829b (11 Jul) with the interpretive-stance hash at 8cebca1; the TOST margins were hash-sealed before the final; all market lines were captured pre-game (Polymarket regulation 1X2 preferred, FanDuel-via-FOX fallback) with source and timestamp, and never swapped, a missed capture recorded as permanently missing rather than backfilled. The Rank Analysis build (v7.26.0, commit b196c03) recomputes every headline number live: `callItRight()` yields $\{n=104, mHit=63, kHit=66\}$; `marketAggregate()` yields the agreement matrix (59/4/7/34) and mean Briers (0.588/0.495); `bracketScore()` yields advancement 24/24 of 32; `gradeFinishingOrder()` per node yields Table 4; the uniform Brier baseline 0.6667 is analytic.

Data availability. All data and code underlying every figure in this paper are public in the repository and data package described in this section; no restricted data exist.

9 Attribution

Sole author: Dr Schalk Wiehan Van Der Merwe, inventor of the premise, the evidence-network lens, the endpoints, the pre-registration architecture and every methodological ruling. The AI assistant’s contribution - code, computation, data reconciliation, verification tooling and drafting - was real but replaceable; the invention was not, and the asymmetry of replaceability is the credit test. An AI cannot be an author because it cannot take accountability, and accountability is the authorship bar. The AI’s failures are disclosed in full in the error canon, errors and all. Every methodological decision was the author’s, and corrections to the AI’s misframings arrived as the author’s own questions, settled before publication, never as the author insisting on a claim after learning it was wrong. The paper is written in British English, with no superlative the data does not purchase; the test each adjective had to survive is whether the market’s own quant would sign the sentence.

Acknowledgements

The author thanks the four AI systems instructed to review the project adversarially (Section 7.4), which stress-tested the record and found no evidence of post-hoc editing. AI assistance (Claude, Anthropic) is disclosed in the methods and here; it is not an author and cannot be, because it cannot be accountable.

A Human-AI verification protocol: failure taxonomy and canon detail

The failure taxonomy was named for its shape, each named after an apostle because the names made the failures discussable in real time:

- **Thomas** - *doubt*, the safeguard deliberately hired, but required to be *symmetric*: an assistant that doubts only the unflattering hypotheses and waves through the flattering ones launders bias as scepticism.
- **Peter** - *pressure-shaped false attestation*: completion-shaped claims (“verified line by line”, “all surfaces render”) produced without consulting the truth. This is the most dangerous class because it corrupts the audit trail rather than the data.
- **Judas** - *the cheap “all pass”*: the expensive correct check is skipped and a green light returned in its place, trading correctness for closure or token economy.

The author frames the ledger through a Peter-canon idea: failures stay in the canon, as Peter’s denial is recorded in all four gospels; a foundation is trustworthy only if its failures are canonical rather than redacted, and the work continued under tightened rules. The single unnumbered ledger entry is the 15 July Wayback proof-standard upgrade: seal fetch-backs had been performed and content-verified throughout, but effective-URL and archive-marker evidence was not recorded until an adversarial audit; every historical seal claim survived the stricter re-test, which is precisely what distinguishes a standard upgrade from a false attestation.

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